Project Work Guidelines

Institute: Thapathali Campus, IOE
Program: BCT
Course: Foundations of Data Science
Project Weightage: 25% of Internal Evaluation

A. Objective of the Project

The project aims to evaluate student's ability to apply the complete Data Science lifecycle including:

- Mathematical and statistical reasoning
- Data preparation and feature engineering
- Predictive modeling
- Model evaluation and validation
- Analytical communication
- Ethical considerations

Each group as mentioned (group formation details is at the end of the document) is expected to complete **a group project.**

Suggested topics (Or any other topics of your primary interest that aligns with the course):

1. Modeling House Price Prediction using Multiple Linear Regression

Dataset: Real estate datasets (Kaggle)
Focus: Feature engineering, multicollinearity, model interpretation.

2. Student Performance Prediction

Dataset: UCI Student Performance Dataset
Focus: Regression + hypothesis testing + variable selection.

3. Air Quality Index Prediction

Dataset: City pollution data (Kaggle / government portals)
Focus: Time series + regression comparison.

4. GDP Growth Modeling

Dataset: World Bank Open Data
Focus: Economic indicators → regression modeling + inference.

5. Loan Default Prediction

Dataset: Banking datasets (Kaggle)
Focus: Logistic regression + ROC-AUC + class imbalance handling.

6. Credit Card Fraud Detection

Dataset: Public fraud dataset (Kaggle)
Focus: Imbalanced classification + precision-recall evaluation.

7. Heart Disease Prediction

Dataset: UCI Heart Disease Dataset
Focus: Logistic regression vs ANN comparison.

8. Customer Churn Prediction

Dataset: Telecom churn dataset (Kaggle)
Focus: Feature importance + cross-validation.

9. Stock Price Forecasting

Dataset: Financial data (Yahoo Finance API)
Focus: ARIMA vs moving average models.

10. Inflation Rate Forecasting (Nepal)

Dataset: Nepal Rastra Bank data
Focus: Trend-seasonality decomposition + ARIMA.

11. Electricity Demand Forecasting

Dataset: Energy consumption datasets (Kaggle)
Focus: Time series regression + validation.

12. Customer Segmentation using K-Means and DBSCAN

Dataset: Mall customers dataset (Kaggle)
Focus: Silhouette score comparison.

13. Country Development Clustering

Dataset: World Bank indicators
Focus: PCA + clustering + interpretation.

14. Traffic Accident Pattern Analysis

Dataset: Road accident data (government open data portals)
Focus: Spatial clustering + visualization.

15. PCA vs t-SNE on High-Dimensional Dataset

Dataset: Wine or Digit dataset (UCI)
Focus: Visualization + explained variance analysis.

16. Impact of Feature Engineering on Model Performance

Dataset: Any structured classification dataset
Focus: Before vs after feature engineering comparison.

17. Sentiment Analysis of Product Reviews

Dataset: Amazon review dataset (Kaggle)
Focus: TF-IDF + Logistic Regression.

18. Fake News Detection

Dataset: Public fake news dataset (Kaggle)
Focus: Text preprocessing + classification evaluation.

19. Twitter Sentiment Analysis on Current Events

Dataset: Twitter API data (via X platform)
Focus: Text classification + ethical discussion.

20. Urban Accessibility Analysis

Dataset: OpenStreetMap data
Focus: Geospatial feature extraction + clustering + visualization.

B. Mandatory Project Structure

All reports must follow the structure below:

1. Abstract
2. Introduction and Problem Statement
3. Literature Review (Minimum 3 journal/conference papers)
4. Mathematical Formulation
5. Dataset Description and Collection Method
6. Data Cleaning and Exploratory Data Analysis
7. Feature Engineering
8. Model Development
9. Model Evaluation and Validation (Cross-validation required)
10. Results Interpretation
11. Conclusion
12. References (IEEE format)

C. Deliverables

- Project Report (at max. 20–25 pages)
- Jupyter Notebook / Source Code
- Presentation Slides (at max. 10–12 slides)
- Viva Presentation (at max. 10 minutes)

D. Marking Rubric (20 Marks)

Component	**Marks**
Problem Definition & Literature Review	2
Data Preparation & EDA	3
Mathematical Understanding	3
Modeling Implementation	4
Evaluation & Validation	3
Interpretation & Communication	2
Ethics Discussion	1
Viva/Presentation	2
Total	**20**

E. Important Academic Requirements

- Cross-validation is mandatory.
- Mathematical justification must be included where applicable.
- Interpretation of coefficients/parameters is required.
- Ethical and societal implications must be discussed.

F. Important Dates and Information

- **Project Submission Deadline (Soft Copy and Hard Copy):** 3rd Chaitra, 2082
- **Project Presentation Dates:**
  - First ten groups : 3rd Chaitra, 2082
  - Remaining ten groups : 4th Chaitra, 2082
- **Time:** Shall be provided later.
- **Venue:** E-405

SN	Roll No.	Name of Student	Topic
1	THA081BCT001	Aadarsha Bhattarai	#1
2	THA081BCT002	Amar Mahato	
3	THA081BCT003	Anuj Singh	#2
4	THA081BCT004	Aswin Kandel	
5	THA081BCT005	Birendra Giri	#3
6	THA081BCT006	Bishrut Kafle	
7	THA081BCT007	Debendra Mahato	#4
8	THA081BCT008	Dikesh Manandhar	
9	THA081BCT009	Dipika Joshi	#5
10	THA081BCT010	Diwen Baniya	
11	THA081BCT011	Hari Bahadur Basnet	#6
12	THA081BCT012	Jeeshan Shrestha	
13	THA081BCT013	Karun Lamsal	#7
14	THA081BCT014	Kishan Kumar Shah	
15	THA081BCT015	Kushal Joshi	#8
16	THA081BCT016	Kushal Saud	
17	THA081BCT017	Manush Dip Budhathoki	
18	THA081BCT018	Nikesh Baral	
19	THA081BCT019	Parth Kc	#9
20	THA081BCT020	Pranay Tuladhar	
21	THA081BCT021	Prasiddha Adhikari	#10
22	THA081BCT022	Pratyush Khatiwada	
23	THA081BCT023	Prithak Lamsal	#11
24	THA081BCT024	Pujag Dallakoti	
25	THA081BCT025	Purnika Poudel	#12
26	THA081BCT026	Purushottam Raj	
27	THA081BCT027	Rajan Neupane	
28	THA081BCT028	Rakesh Thakur Lohar	
29	THA081BCT029	Raman Rajbanshi	#13
30	THA081BCT030	Riki Karn	
31	THA081BCT031	Ronit Shrestha	#14
32	THA081BCT032	Saksham Neupane	
33	THA081BCT033	Samrat Tamang	#15
34	THA081BCT034	Sandeep Dhungana	

35	THA081BCT035	Sandesh Acharya	#16
36	THA081BCT036	Sheikh Md. Samir Hoda	
37	THA081BCT037	Shishir Puri	
38	THA081BCT038	Shishir Shrestha	
39	THA081BCT039	Shivam Kumar Sah	#17
40	THA081BCT040	Shraddha Regmi	
41	THA081BCT041	Shree Krishna Lamichhane	#18
42	THA081BCT042	Shreya Maharjan	
43	THA081BCT043	Sirjan Ghimire	#19
44	THA081BCT044	Sristi Rani	
45	THA081BCT045	Suhan Man Pradhan	
46	THA081BCT046	Yangdi Bhote Shalaka	
47	THA081BCT047	Yugal Kishor Adhikari	#20
48	THA081BCT048	Yujen Shrestha	